

Abstract

Proteins interact with many molecules while they exist. Some of these molecules can bind to proteins, potentially altering their function. In this poster, we present EProM—a visual analysis interface to explore protein modifications—as a contribution to the IEEE VIS 2022 Bio+MedVis Challenge.

Protein structures are described in three levels:

- Primary structure: describes the amino acid sequence,
- Secondary structure: describes connections between amino acids:
 - α -helix: connections between consecutive amino acids forming a helix,
 - β -strand: connections between non-consecutive amino acids that are part of a helix forming a β -sheet,
 - Unstructured: no additional connections,
- Tertiary structure: describes the 3D arrangement.

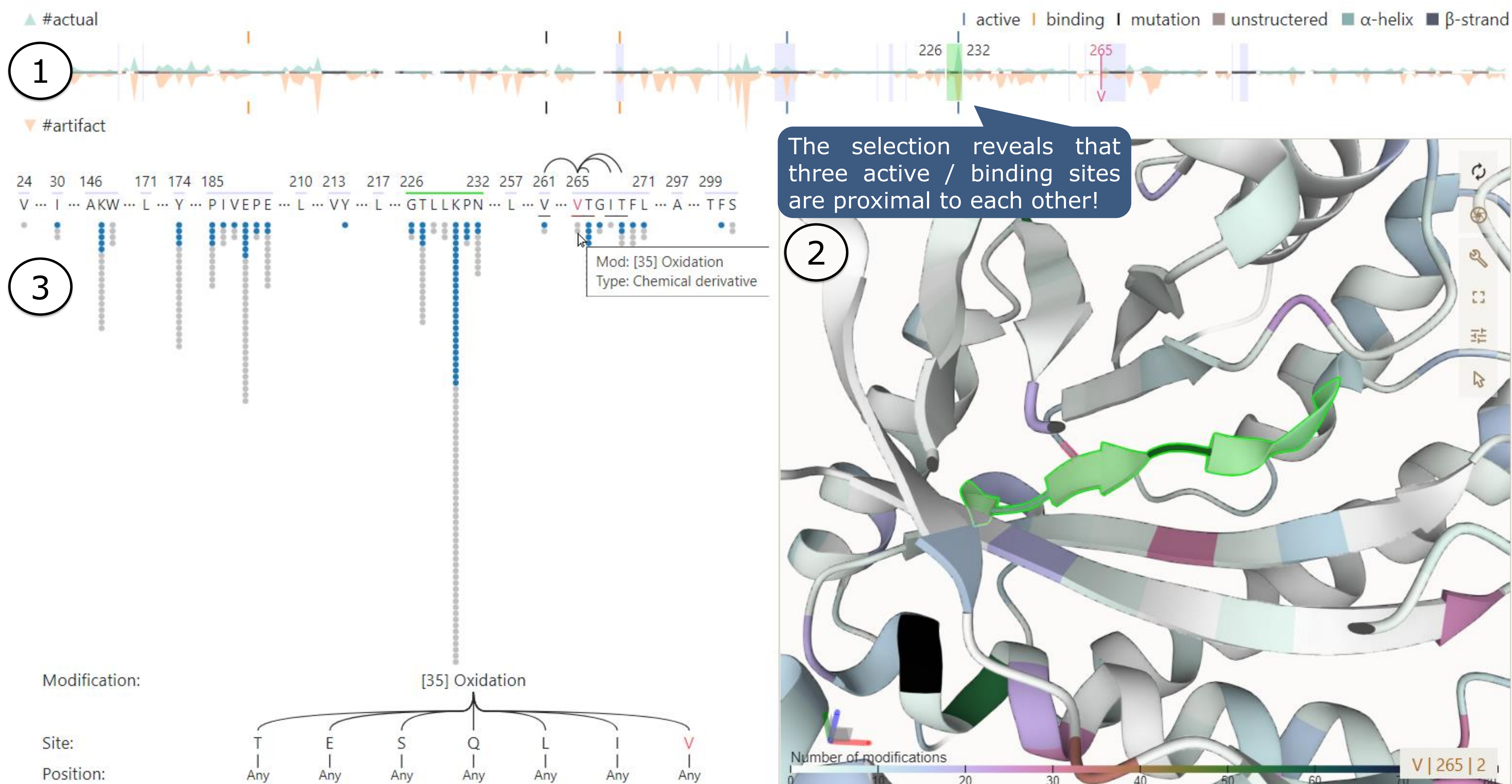


Figure 1. Main EProM visual analytics interface consisting of (1) a browser view used to navigate through the protein's primary structure; (2) a Mol* viewer [1] showing the number of modifications in the protein's 3D structure; and (3) a detail plot visualizing modifications of interest measured in the selected area.

Design and Implementation

The *browser view* (Fig 1.1) visualizes the number of modifications along the primary structure, distinguishing artefacts from actual modifications. Furthermore, the secondary structure—unstructured, α -helices, and β -strands—and residues related to pathogenic mutations, active sites, and binding sites are annotated along the x-axis.

Brushing in the browser view will automatically focus the *Mol* viewer* [1] (Fig 1.2) and *detail plot* (Fig 1.3) on the selected residues.

The *detail plot* (Fig 1.3) visualizes all modifications observed on the selected residues and residues proximal to the selection, their possible binding sites, and measurement uncertainty. Hovering over a residue highlights it in red in all three views.

Acknowledgements

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References

[1] D. Sehnal, S. Bittrich, M. Deshpande, R. Svobodová, K. Berka, V. Bazgier, S. Velankar, S. K. Burley, J. Koča, and A. S. Rose. Mol* Viewer: modern web app for 3D visualization and analysis of large biomolecular structures. Nucleic Acids Research, 49(W1):W431–W437, 05 2021. doi: 10.1093/nar/gkab314

Redesign

Our redesign (Fig 2) was developed with the objective to provide an intuitive overview of the possible modifications for the entire protein and per residue. We settled on a polar plot that shows all modifications as circles colored by their classification, placing modification names along the circumference and mapping residue positions to the radius.

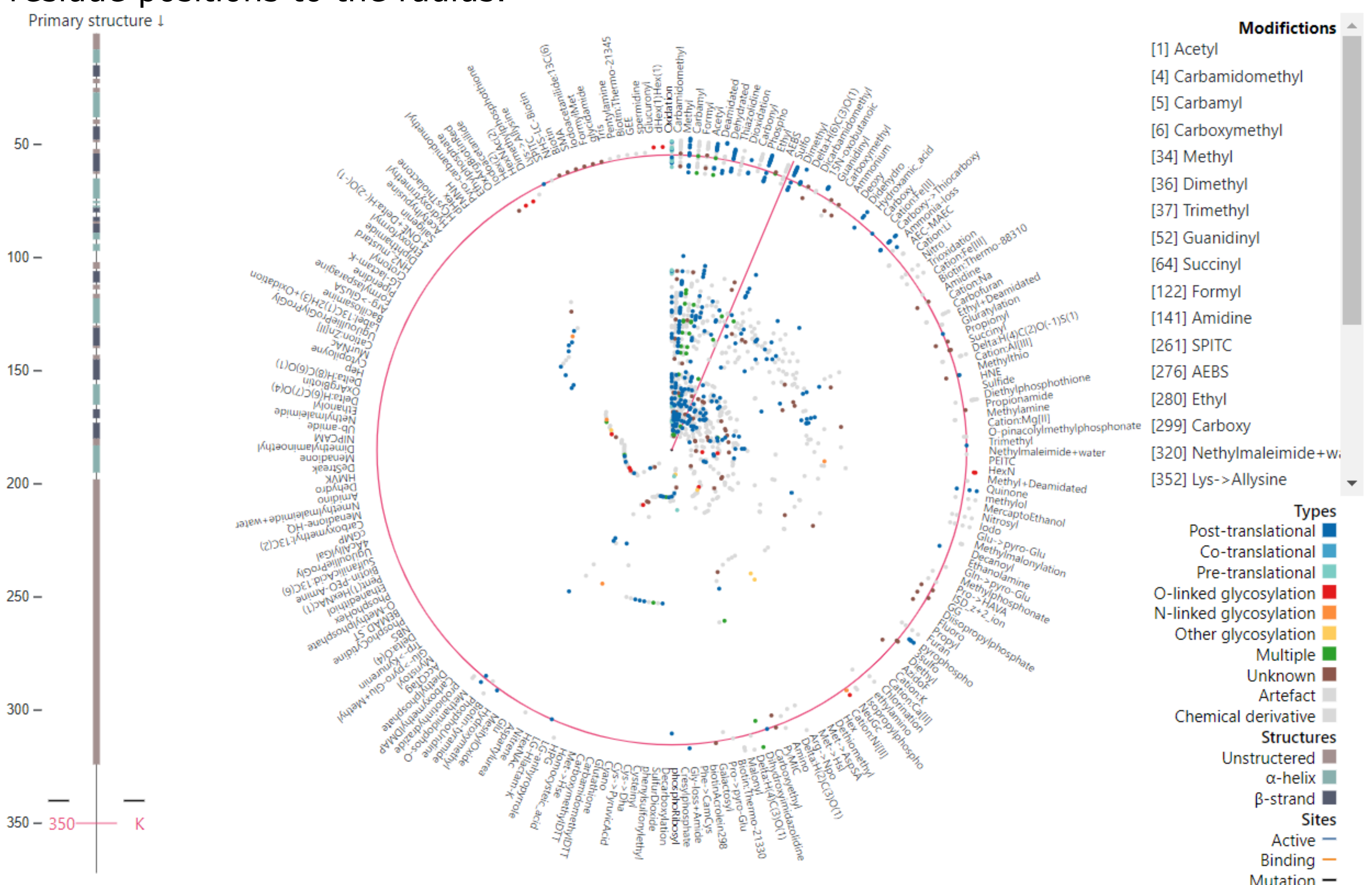


Figure 2. Redesign of the existing 2D visualization. Modifications are shown as dots in a polar plot and colored by their classification. The red circle indicates the hovered residue's modifications, and the red line indicates the mouse position on the radial modification scale.